

Chapter 19 Trilobites, Chelicerates, and Myriapods

Zoology

Instructor

November 28, 2025

Outline

Phylum Arthropoda (节肢动物门)

Subphylum Trilobita (三叶虫亚门)

Subphylum Chelicerata (螯肢亚门)

Class Arachnida (蛛形纲)

Subphylum Myriapoda (多足亚门)

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Characteristics of Arthropods (1)

- Most **species-diverse** phylum (3/4 of all known species).
- Occur in all types of environments.
- Exoskeleton (外骨骼): Versatile cuticle (角质层) secreted by epidermis.
 - Inner thick **procuticle** (原角质层) (exocuticle + endocuticle).
 - Outer thin **epicuticle** (上角质层) (protein + lipids).
 - Contains **chitin** (几丁质) (nitrogenous polysaccharide).
 - Sclerotization (硬化) and calcification (in crustaceans).
- Molting (Ecdysis) (蜕皮): Shedding of exoskeleton for growth.

Structure of Cuticle

- **Epicuticle** (上角质层): Thin, waxy, protective.
- **Procuticle** (原角质层): Thick, chitinous, provides strength.
- **Epidermis** (表皮): Secretes the cuticle.



Figure 1: Structure of crustacean cuticle.

Characteristics of Arthropods (2)

- Segmentation and Appendages (分节与附肢):
 - **Tagmata** (体部): Functional groups of segments (e.g., head, thorax, abdomen).
 - **Appendages** (附肢): Jointed levers.
 - **Uniramous** (单肢型): Single branch (e.g., insects).
 - **Biramous** (双肢型): Two branches (e.g., crustaceans).
- Mouthparts (口器):
 - **Chelicerae** (螯肢): No mandibles (e.g., spiders).
 - **Mandibles** (大颚): Jaws (e.g., insects).

Arthropod Appendages and Mouthparts

- Comparison of **uniramous** (单肢型) vs. **biramous** (双肢型) appendages.
- Comparison of **chelicerae** (螯肢) vs. **mandibles** (大颚).



Figure 2: Two important arthropod characters: Appendages may be uniramous (honey bee leg) or biramous (lobster limbs); mouthparts may include chelicerae (spider) or mandibles (grasshopper). Note that presence or absence of gills is unrelated to appendage form.

Characteristics of Arthropods (3)

- Respiration (呼吸):
 - Terrestrial: **Tracheal system** (气管系统) or **book lungs** (书肺).
 - Aquatic: **Gills** (鳃).
- Sensory Organs (感觉器官): Highly developed (compound eyes, chemical reception).
- Behavior: Complex innate and learned behaviors.
- Metamorphosis (变态): Reduces competition between stages.

Phylogeny of Arthropods

- Clade **Panarthropoda** (泛节肢动物).
- Five Subphyla:
 - **Trilobita** (三叶虫亚门) (Extinct).
 - **Chelicerata** (螯肢亚门) (Spiders, horseshoe crabs).
 - **Myriapoda** (多足亚门) (Centipedes, millipedes).
 - **Crustacea** (甲壳亚门) (Lobsters, crabs).
 - **Hexapoda** (六足亚门) (Insects).
- **Pancrustacea** (泛甲壳动物):
Crustacea + Hexapoda.



Figure 3: Cladogram of arthropods showing probable relationships of the four extant subphyla. Only a few synapomorphies are included here. Crustaceans and hexapods are shown as sister taxa, but no branching order for Myriapoda, Chelicerata, or Pancrustacea is specified.

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Subphylum Trilobita

- Extinct for 200 million years (flourished in Cambrian/Ordovician).
- Body: Trilobed (two longitudinal grooves).
- Tagmata: **Cephalon** (头), **Trunk** (躯干), **Pygidium** (尾板).
- Appendages: **Biramous** (双肢型) on each segment except the last.
- Bottom dwellers, scavengers.

Trilobite and Eurypterid Fossils

- A: **Trilobite** fossils (dorsal view).
- B: **Eurypterid** fossil (giant water scorpion).



Figure 4: Fossils of early arthropods. A, Trilobite fossils, dorsal view. These animals were abundant in mid-Cambrian period. B, Eurypterid fossil. Eurypterids flourished in Europe and North America from Ordovician to Permian periods.

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Subphylum Chelicerata Characteristics

- Tagmata: Two.
 - **Cephalothorax** (prosoma) (头胸部/前体).
 - **Abdomen** (opisthosoma) (腹部/后体).
- Appendages: Six pairs on cephalothorax.
 - 1 pair of **chelicerae** (螯肢) (mouthparts).
 - 1 pair of **pedipalps** (脚须).
 - 4 pairs of **walking legs** (步足).
- No antennae (无触角).
- Most suck liquid food.

Chelicerate Phylogeny

- Shows relationships among Merostomata, Pycnogonida, and Arachnida.



Figure 5: Cladogram of chelicerates, showing one proposed ancestor-descendant relationship within the chelicerate clade.

Class Merostomata (肢口纲)

- Subclass Eurypterida (板足鲎亚纲): Extinct giant water scorpions.
- Subclass Xiphosurida (剑尾亚纲) (Horseshoe Crabs):
 - “Living fossils” (e.g., **Limulus**).
 - **Carapace** (背甲): Unsegmented, horseshoe-shaped.
 - **Telson** (尾节): Long tailpiece.
 - **Book gills** (书鳃): Flat, leaf-like gills.

Horseshoe Crab (**Limulus**)

- A: Dorsal view showing carapace and telson.
- B: Ventral view showing appendages and book gills.



Figure 6: A, Dorsal view of horseshoe crab **Limulus** (class Merostomata). They grow to 0.5 m in length. B, Ventral view of female.

Class Pycnogonida (海蜘蛛纲)

- Small, thin bodies.
- Usually 4 pairs of long walking legs.
- **Ovigers** (抱卵肢): Subsidiary legs in males for carrying eggs.
- **Proboscis** (吻): Long, suctorial.
- Reduced abdomen.

Sea Spiders

- A: **Nymphon** sp.
- B: **Pycnogonum hancockii**.



Figure 7: A, Pycnogonid, **Nymphon** sp. In this genus all anterior appendages (chelicerae, palps, and ovigers) are present in both sexes, although ovigers are often not present in females of other genera. B, **Pycnogonum hancockii**, a pycnogonid with relatively short legs. Females of this genus have neither chelicerae nor ovigers and males have ovigers.

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Class Arachnida Overview

- Includes spiders, scorpions, ticks, mites, harvestmen.
- Tagmata: Cephalothorax and Abdomen.
- Predaceous (mostly).
- Modifications: Fangs (chelicerae), claws (pedipalps), venom glands, stingers.

Order Araneae: Spiders (蜘蛛目)

- Body: Compact; unsegmented cephalothorax and abdomen joined by **pedicel** (腹柄).
- Appendages:
 - **Chelicerae** (螯肢) with terminal fangs (venom).
 - **Pedipalps** (脚须) (sensory, sperm transfer).
 - 4 pairs of walking legs.

Spider External Anatomy

- Jumping spider anatomy.
- Note chelicerae, pedipalps, and eyes.



Figure 8: External anatomy of a jumping spider, with anterior view of head (at right).

Spider Internal Anatomy

- Respiration: **Book lungs** (书肺) and/or **tracheae** (气管).
- Excretion: **Malpighian tubules** (马氏管) (uric acid) and **coxal glands** (基节腺).
- Sensory: Simple eyes (8 usually), sensory setae.
- Silk: Silk glands and **spinnerets** (纺绩器).

Spider Internal Structure

- Shows digestive, circulatory, respiratory, and reproductive systems.



Figure 9: Spider, internal anatomy.

Predation and Webs

- Spiders are predaceous.
- Use silk for webs, wrapping prey, egg sacs, etc.
- Some chase prey (Wolf spiders, Jumping spiders).

Prey Capture in Web

- Grasshopper snared in **Argiope** web.
- Wrapped in silk.



Figure 10: Grasshopper, snared and helpless in the web of a golden garden spider (**Argiope aurantia**), is wrapped in silk while still alive. If the spider is not hungry, the prize will be saved for a later meal.

Hunting Strategies

- Not all spiders use webs to capture prey.
- Some are active hunters.
- Examples: Wolf spiders, Jumping spiders, Fisher spiders.
- They chase or ambush prey directly.

Active Hunting

- Fisher spider (**Dolomedes**) capturing a minnow.
- Does not use a web for capture.



Figure 11: Fisher spider, **Dolomedes triton**, feeds on a minnow. This handsome spider feeds mostly on aquatic and terrestrial insects but occasionally captures small fishes and tadpoles. It pulls its paralyzed victim from the water, pumps in digestive enzymes, then sucks out the predigested contents.

Venomous Spiders

- Most spiders are harmless to humans.
- Black Widow (**Latrodectus**): Neurotoxic venom.
- Brown Recluse (**Loxosceles**): Hemolytic venom (tissue necrosis).
- Tarantulas: Large but rarely dangerous.

Tarantula

- **Brachypelma vagans.**
- Large, hairy, mostly harmless.



Figure 12: A tarantula, **Brachypelma vagans.**

Dangerous Species

- While tarantulas are large, they are usually not dangerous.
- The most dangerous spiders in the US are smaller.
- **Latrodectus** (Black Widow) and **Loxosceles** (Brown Recluse).
- Their venoms have different mechanisms of action.

Dangerous Spiders

- A: Black widow (***Latrodectus***).
- B: Brown recluse (***Loxosceles***).



Figure 13: A, Black widow spider, ***Latrodectus mactans***, suspended on her web. Note the red “hourglass” on the ventral side of her abdomen. B, Brown recluse spider, ***Loxosceles reclusa***, is a small venomous spider. Note the small violin-shaped marking on its cephalothorax. The venom is hemolytic and dangerous.

Order Scorpiones & Opiliones

- Scorpiones (Scorpions) (蝎目):
 - Short cephalothorax, **preabdomen** (前腹部), **postabdomen** (后腹部) with **stinger** (螫针).
 - Large chelate pedipalps.
 - Viviparous.
- Opiliones (Harvestmen) (盲蛛目):
 - Abdomen and cephalothorax broadly joined (no pedicel).
 - Long, spindly legs.
 - Scavengers, not venomous.

Scorpion and Harvestman

- A: Emperor scorpion (**Paninus**) with young.
- B: Harvestman (**Mitopus**).



Figure 14: A, An emperor scorpion (order Scorpiones), **Paninus imperator**, with young, which stay with the mother until their first molt. B, Harvestmen, **Mitopus** sp. (order Opiliones). Harvestmen run rapidly on their stiltlike legs. They are especially noticeable during the harvesting season, hence the common name.

Order Acari: Ticks and Mites (蜱螨目)

- Medically and economically important.
- Body: Complete fusion of cephalothorax and abdomen.
- Capitulum (颚体): Anterior projection with mouthparts.
- Many are parasitic or pests.

Ticks and Mites

- A: Wood tick (**Dermacentor**).
- B: Red velvet mite (**Trombidium**).



Figure 15: A, Wood tick, **Dermacentor variabilis** (order Acari). Larvae, nymphs, and adults are all ectoparasites or micropredators that drop off their hosts to molt to the next stage. B, Red velvet (harvest) mite, **Trombidium** sp. As with chiggers (**Trombicula**), only larvae of **Trombidium** are ectoparasites. Nymphs and adults are free-living and feed on insect eggs and small invertebrates.

Mites in the Home

- Mites are found in many habitats, including human dwellings.
- House dust mites feed on organic detritus.
- Their feces can cause allergic reactions in humans.

House Dust Mite

- **Dermatophagoides farinae.**
- Causes allergies.



Figure 16: Scanning electron micrograph of house dust mite, **Dermatophagoides farinae.**

Mites in Agriculture

- Many mites are serious pests for crops.
- Spider mites (family Tetranychidae) are a prime example.
- They pierce plant cells and suck out the contents.
- Can cause significant economic damage.

Agricultural Pests

- Spider mites (Tetranychidae).
- Damage plants by sucking cell contents.



Figure 17: Damage to *Chamaedorea* sp. palm caused by mites of the family Tetranychidae (order Acari). Over 130 species of this family occur in North America, and some are serious agricultural pests. Mites pierce plant cells and suck out contents, giving leaves the mottled appearance shown here.

Mites on Humans

- Some mites live on the human body without causing harm.
- **Demodex** species live in hair follicles and sebaceous glands.
- They are present in most adults.

Human Parasites (1)

- **Demodex folliculorum** (Human follicle mite).
- Usually non-pathogenic.



Figure 18: **Demodex folliculorum**, human follicle mite.

Pathogenic Mites

- Some mites cause skin diseases.
- **Sarcoptes scabiei** burrows into the skin.
- Causes intense itching (scabies).
- Highly contagious.

Human Parasites (2)

- **Sarcoptes scabiei** (Human itch mite).
- Causes scabies (burrows in skin).



Figure 19: **Sarcoptes scabiei**, human itch mite.

Ticks and Disease

- Ticks are major vectors of disease.
- Second only to mosquitoes in disease transmission.
- Transmit bacteria, viruses, and protozoa.
- Examples: Lyme disease, Texas cattle fever.

Disease Vectors

- **Boophilus annulatus** (Cattle tick).
- Transmits Texas cattle fever.
- Ticks also transmit Lyme disease (**Ixodes**) and Rocky Mountain spotted fever (**Dermacentor**).



Figure 20: **Boophilus annulatus**, a tick that carries Texas cattle fever.

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Subphylum Myriapoda Characteristics

- “Many footed”.
- Tagmata: Head and Trunk.
- Appendages: Paired, **uniramous** (单肢型) on most trunk segments.
- Respiration: Tracheae (气管).
- Excretion: Malpighian tubules (马氏管).

Myriapod Phylogeny

- Relationships of Chilopoda, Diplopoda, Pauropoda, and Symphyla.



Figure 21: Cladogram showing hypothetical relationships of myriapods. Organs of Tömösvary are unique sensory organs opening at the bases of the antennae, and repugnatorial glands, located on certain segments or legs, secrete an obnoxious substance for defense. The gnathochilarium is formed in diplopods and pauropods by fusion of the first maxillae, and the collum is the collarlike tergite of the first trunk segment.

Class Chilopoda (Centipedes) (唇足纲)

- Flattened bodies.
- One pair of legs per segment.
- Venom claws (毒爪/颚足) (maxillipeds) on first segment.
- Active predators.

Centipede

- A: **Scolopendra**.
- B: Head structure.



Figure 22: A, A centipede, **Scolopendra** (class Chilopoda) from the Amazon Basin, Peru. Most segments have one pair of appendages each. First segment bears a pair of venom claws, which in some species can inflict serious wounds. Centipedes are carnivorous. B, Head of centipede.

Class Diplopoda (Millipedes) (倍足纲)

- Cylindrical bodies.
- Two pairs of legs per abdominal segment (**diplosegments** - 双体节).
- Herbivorous or scavengers.
- **Repugnatorial glands** (防御腺) for defense.

Millipede

- A: Tropical millipede.
- B: Head structure.



Figure 23: A, A tropical millipede with warning coloration. Note the typical doubling of appendages on most segments, hence diplosegments. B, Head of millipede.

Classes Pauropoda and Symphyla

- Pauropoda (少足纲): Minute, soft-bodied, branched antennae, no true eyes.
- Symphyla (综合纲): Small, centipede-like, live in humus.

Pauropod and Symphylan

- A: Pauropod.
- B: **Scutigera** (Symphylan).



Figure 24: A, Pauropod. Pauropods are minute, whitish myriapods with three-branched antennae and nine pairs of legs. They live in leaf litter and under stones. They are eyeless but have sense organs that resemble eyes. B, **Scutigera**, a symphylan, is a minute whitish myriapod that is sometimes a greenhouse pest.